MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
Discipline Specific Elective (DSE) – MG5DSECHE303 – NANOTECHNOLOGY FOR ENERGY APPLICATIONS
MODEL QUESTION PAPER
(2024 ADMISSION ONWARDS)

Time: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

1.	What is meant by global energy consumption? Mention one factor affecting it.	[U | CO1]
2.	Distinguish between renewable and non-renewable energy sources with one example each.	[An | CO1]
3.	State one advantage and one limitation of conventional energy sources.	[U | CO1]
4.	What is meant by intermittency in renewable energy systems?	[K | CO1]
5.	What is a photovoltaic cell?	[K | CO2]
6.	Mention one important component of a dye-sensitized solar cell (DSSC).	[U | CO2]
7.	What is meant by a bulk heterojunction solar cell?	[U | CO2]
8.	What is the function of the cathode in a lithium-ion battery?	[U | CO3]
9.	What is an electrical double layer in a capacitor?	[U | CO3]
10.	Mention two materials used for hydrogen storage.	[K | CO3]

11.	What are metal–organic frameworks (MOFs)?	[U | CO4]
12.	Name two 2D materials used in energy-related applications.	[K | CO4]

(2 × 10 = 20)

Part B

Short Answer Questions

Answer any 6 questions

Each question carries 5 marks

13.	Analyse the major advantages and limitations of conventional energy sources with suitable examples.	[An | CO1]
14.	Discuss the major technological and economic challenges involved in the large-scale implementation of renewable energy technologies.	[An | CO1]
15.	Explain the process of tidal, geothermal and biomass energy conversion.	[U | CO1]
16.	Explain the principle and architecture of a photovoltaic cell. How is solar energy converted into electrical energy?	[U | CO2]
17.	Compare DSSC and quantum-dot solar cells based on their operating principles and materials.	[An | CO2]
18.	Explain the working principle and architecture of a lithium-ion battery. Mention the functions of its major components.	[U | CO3]
19.	Differentiate between EDLCs and pseudocapacitors based on their charge-storage mechanisms.	[U | CO3]
20.	Explain the important materials and methods used for hydrogen storage, with special reference to MOFs and metal hydrides.	[U | CO3]

(5 × 6 = 30)

Part C

Essay Questions

Answer any 2 questions

Each question carries 10 marks

21.	Analyse the present global energy scenario with reference to: a) global energy requirements and consumption b) challenges in renewable-energy implementation,	[An | CO1]

22. Explain the different photovoltaic technologies, including:
    a) perovskite solar cells [U | CO2]
    b) silicon solar cells.

23. Discuss the principles and materials of energy-storage technologies with reference to:
    a) primary and secondary batteries [U | CO3]
    b) lithium-ion batteries and electrode materials
    c) hydrogen-storage materials.

24. Discuss the role of advanced nanostructured materials in energy storage and conversion, with reference to:
    a) MOFs and COFs [An | CO4]
    b) graphene, boron nitride and carbon nitride
    c) two-dimensional metal chalcogenides such as MoS_2 and $MoSe_2$.

(10 × 2 = 20)
